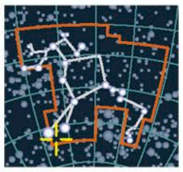
MAIN-SEQUENCE STARS

During a star's life, it passes through many phases, but most of its time will be spent on the main sequence. This means that the chances of seeing any star are greatest during its main-sequence lifetime. In fact, about 90 percent of all observed stars are on the main sequence. Although main-sequence stars are spread throughout the Milky Way, they appear predominantly in its plane and central bulge.

PROMINENT STARS
Known as the Pointers, Alpha and Beta Centauri are prominent main-sequence stars guiding the way to the Southern Cross.

ORANGE-RED STAR

Proxima Centauri



CENTAURUS

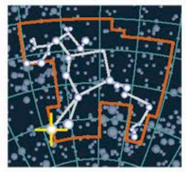
DISTANCE FROM SUN 4.2 light-years
MAGNITUDE 11.05
SPECTRAL TYPE M

The closest star to the Sun, Proxima Centauri is far too faint to be seen with the naked eye—as a result, it was only discovered in 1915. A red dwarf star with just one-tenth the mass of the Sun, it is thought to be an outlying member of the Alpha Centauri system (see right), orbiting the central stellar pair at a distance of 10,000 au, with a period of at least 1 million years. Proxima is also a flare star—despite its faintness, it periodically emits huge bursts of radiation that see it brighten by a whole magnitude (see pp.282–283). Even during these eruptions, however, it remains 18,000 times dimmer than the Sun—despite becoming an intense source of low-energy X-rays and high-energy ultraviolet. In 2016, tiny shifts in the wavelength of Proxima's light (see p.297) revealed the presence of a roughly Earth-sized planet orbiting in Proxima's habitable zone where water might survive on the surface.

FLARING DWARF
Despite its general faintness, Proxima's harsh radiation outbursts make life on its orbiting planet extremely unlikely.

YELLOW AND ORANGE STARS

Alpha Centauri



CENTAURUS

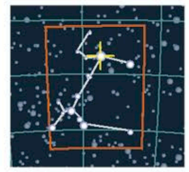
DISTANCE FROM SUN 4.3 light-years
MAGNITUDES 0.0 and 1.3
SPECTRAL TYPES G and K

The two stars of Alpha Centauri—also known as Rigil Kentaurus—orbit each other every 79.9 years. They are very close and, in some images (below), are distinguishable only by seeing two sets of diffraction spikes. Alpha Centauri A is the brighter and more massive, at 1.57 times the luminosity and 1.1 times the mass of the Sun. Alpha Centauri B is both less massive and less luminous than the Sun.

ALPHA CENTAURI A AND B

WHITE STAR

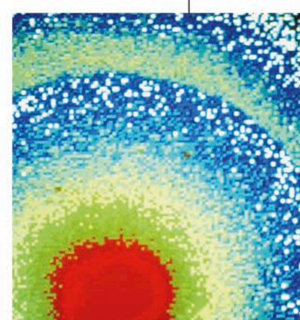
Sirius A



CANIS MAJOR

DISTANCE FROM SUN 8.6 light-years
MAGNITUDE -1.46
SPECTRAL TYPE A

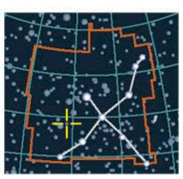
The brightest star in the night sky, Sirius is the ninth-closest star to Earth. It is a binary star, with Sirius A being a main-sequence star and its companion a white dwarf. Sirius A has twice the mass of the Sun and is 23 times as luminous. Recent observations suggest that it may have a stellar wind—the first spectral type A star to show evidence of one.



SCORCHING STAR
A false-color image shows the diffraction pattern of Sirius, the brightest star in the sky. Its name is from the Greek for "scorching."

ORANGE STAR

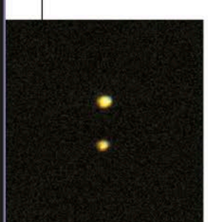
61 Cygni



CYGNUS

DISTANCE FROM SUN 11.4 light-years
MAGNITUDES 5.2 and 6.1
SPECTRAL TYPE K

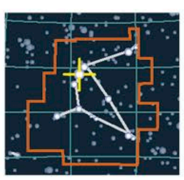
61 Cygni is a binary system of two main-sequence stars that orbit each other every 653 years. It is believed that 61 Cygni has at least one massive planet and possibly as many as three. In 1838, German astronomer Friedrich Bessel became the first to measure the distance of a star from Earth accurately, when he calculated 61 Cygni's annual parallax (see p.70). He chose 61 Cygni because, at that time, it was the star with the largest known proper motion.



FAST STAR

WHITE STAR

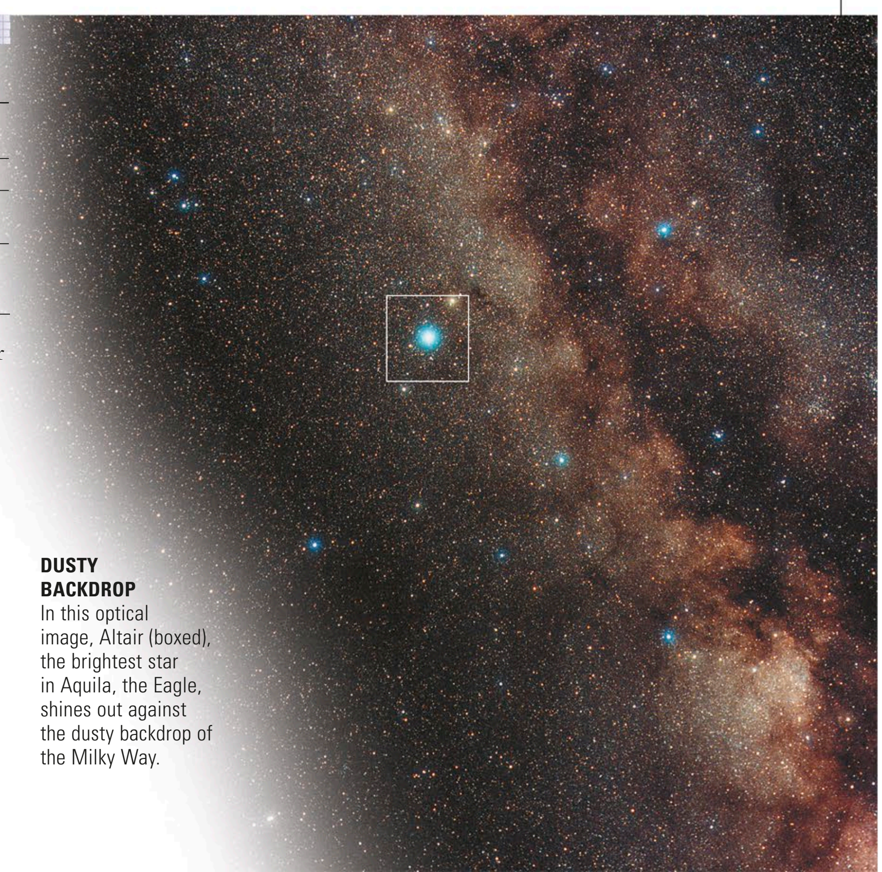
Altair



AQUILA

DISTANCE FROM SUN 16.8 light-years
MAGNITUDE 0.77
SPECTRAL TYPE A

One of the three stars of the Summer Triangle, Altair is the 12th-brightest star in the sky. With a diameter about 1.6 times that of the Sun, it rotates once every 6.5 hours. This puts its equatorial spin rate at about 559,000 mph (900,000 kph), which causes distortion of its overall shape. This distortion is such that the star becomes wider at the equator and flattened at the poles, and estimates have suggested that its equatorial diameter is as much as double its polar diameter. It has a surface temperature of about 17,000°F (9,500°C) and a high rate of proper motion through the Milky Way.



DUSTY BACKDROP
In this optical image, Altair (boxed), the brightest star in Aquila, the Eagle, shines out against the dusty backdrop of the Milky Way.